Question	Answer	Marks
5(a)	$Q = Q_1 = Q_2$ <u>and</u> $V = V_1 + V_2$	M1
	$V = Q / C$ so: $Q / C = Q / C_1 + Q / C_2$ <u>leading to</u> $1 / C = 1 / C_1 + 1 / C_2$	A1
5(b)	total capacitance = $C + ½C = (3 / 2)C$	C1
	total capacitance = gradient = $400 \times 10^{-6} / 6.0$	C1
	either: $C = (2 \times 400 \times 10^{-6}) / (3 \times 6.0) = 4.4 \times 10^{-5}$ F = 44 μF *or*: $C = (2 \times 400) / (3 \times 6.0)$ = 44 μF	A1
5(c)(i)	$\tau = RC$	C1
	$= 54 \times 10^3 \times (3/2) \times 44 \times 10^{-6}$ = 3.6 s	A1
5(c)(ii)	$0.15 = \exp(–t / 3.6)$	C1
	t = 6.8 s	A1

Question	Answer	Marks